BASIC ELECTRONICS ENGINEERING • PART B

UNIT I

Semiconductor Devices

From vacuum tubes to nano-electronics, diodes, Zener diode, BJT and the CE amplifier

WHAT YOU WILL LEARN

1 Evolution of electronics
2 PN junction diode & its characteristics
3 Zener effect & Zener diode
4 Bipolar Junction Transistor (BJT)
5 CB, CE, CC configurations
6 Input / output characteristics
7 Small-signal CE amplifier
8 Formulas, summary & exam questions

1 Evolution of Electronics: Vacuum Tubes to Nano-electronics

Electronics is the branch of engineering that controls the flow of electrons through vacuum, gases and (mostly today) **semiconductors** to process information and power. Every phone, TV and computer is a descendant of one idea: *use a small signal to control a large flow of electrons*.

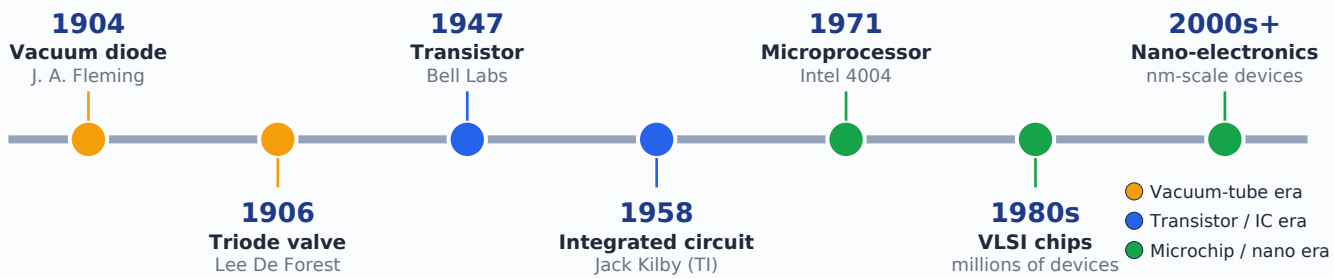


Fig 1.1 Milestones in the evolution of electronics

Vacuum tube: the first electronic device

- A heated **cathode** releases electrons (*thermionic emission*); a positive **anode (plate)** collects them. A **grid** in a triode controls the current, hence it amplifies.
- Used in early radio, TV, radar and the first computers (ENIAC, ~18,000 tubes).

Property	Vacuum tube	Transistor / IC
Size	Large and bulky	Tiny (nm to mm)
Power	High (heater needed)	Very low
Warm-up time	Needs seconds to heat	Instant on
Life and strength	Short life, fragile glass	Very long, rugged
Operating voltage	Hundreds of volts	A few volts
Cost (mass production)	High	Extremely low

Integration levels and nano-electronics

Era	Approx. transistors per chip	Example
SSI / MSI	up to a few hundred	Logic gates, counters
LSI	thousands to ~100,000	Early microprocessors, memories
VLSI / ULSI	lakhs to billions	Modern CPUs, mobile SoCs

- **Nano-electronics** deals with devices whose key dimensions are in the nanometre range (1–100 nm), where **quantum effects** become important.

- Examples: **FinFET / gate-all-around** transistors, **carbon-nanotube** and **graphene** devices, quantum dots, nanowires, memristors.
- Benefits: higher speed, lower power, huge density. Challenges: leakage, heat, tunnelling and fabrication cost.

Applications of nano-electronics
Smartphone processors, AI accelerators, flash memory, medical biosensors, flexible displays, solar cells.

Did you know?
- The ENIAC computer (1946) used about 18,000 vacuum tubes and filled a large hall; a modern phone chip packs **billions** of transistors in a fingernail-sized die.
- A human hair is ~80,000 nm wide, while today's smallest transistor features are only a few nanometres.

2 PN Junction Diode

A **semiconductor** (Si, Ge) conducts better than an insulator but worse than a conductor. Adding impurities (**doping**) makes it *extrinsic*.

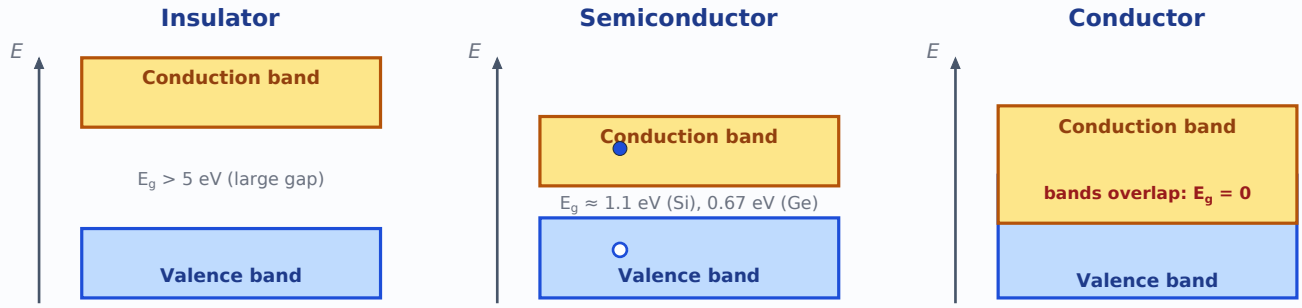


Fig 1.2 Energy-band view: the gap between valence and conduction bands decides how well a material conducts

	N-type	P-type
Dopant	Pentavalent (P, As, Sb): *donor*	Trivalent (B, Al, Ga): *acceptor*
Majority carriers	Free electrons	Holes
Minority carriers	Holes	Electrons
Charge of crystal	Neutral overall	Neutral overall

DEFINITION
A **PN junction diode** is formed by joining P-type and N-type semiconductors. It is a **two-terminal, one-way device**: Anode (P) and Cathode (N). Symbol: ▶| (current flows easily from anode to cathode).

Formation of the depletion region

- Electrons from N-side diffuse across to the P-side; holes diffuse the other way and they **recombine** near the junction.
- This leaves uncovered fixed ions, which form a thin **depletion region** with no free carriers.
- The ions create an internal electric field, the **barrier potential**: about **0.7 V for Si** and **0.3 V for Ge** (at 25 °C).

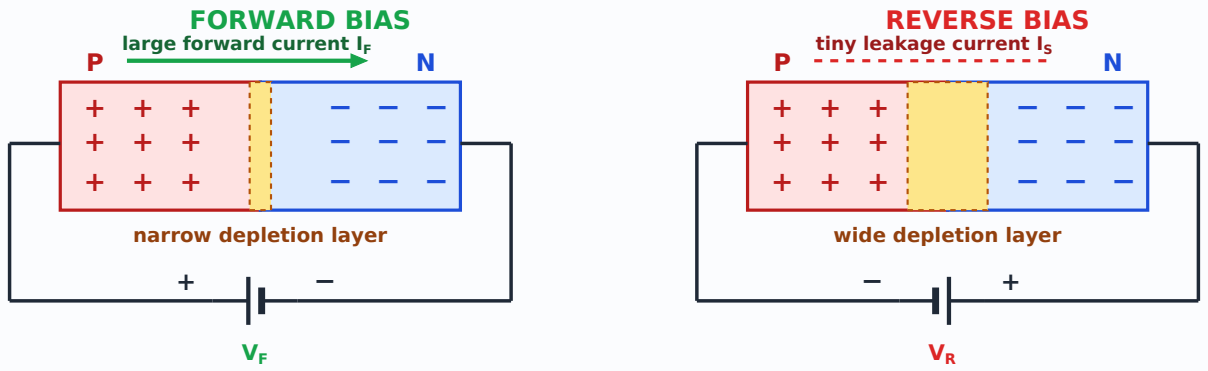


Fig 1.3 PN junction under forward bias (left) and reverse bias (right)

	Forward bias (P to +, N to −)	Reverse bias (P to −, N to +)
Depletion region	Becomes **narrow**	Becomes **wide**
Barrier potential	Reduced	Increased
Current	Large; flows once V > knee voltage	Very small leakage (minority carriers) I_S
Diode acts as	Closed switch (ON)	Open switch (OFF)

3 V-I Characteristics of a PN Junction Diode

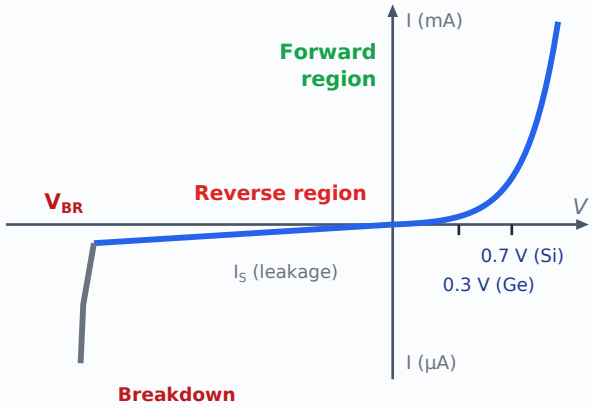


Fig 1.4 V-I characteristic of a PN junction diode (forward and reverse scales differ: mA vs μA)

Diode equation
$$I = I_0 \left(e^{V / \eta V_T} - 1 \right)$$
I_0 = reverse saturation current; $V_T = kT/q \approx 26$ mV at 300 K; η = 1 (Ge), about 2 (Si)

- **Knee (cut-in) voltage:** below it, current is almost zero. Si ≈ 0.7 V, Ge ≈ 0.3 V.
- **Reverse saturation current:** Si in nA, Ge in μA; it rises with temperature (roughly doubles every 10 °C).
- **Breakdown voltage V_{BR}:** a large reverse voltage makes the current rise suddenly and can destroy an ordinary diode.
- **Static resistance** $R_{DC} = V / I$. **Dynamic resistance** $r_d = \Delta V / \Delta I \approx$ **26 mV / I_D** (small in the forward region).

Example: dynamic resistance
A Si diode carries $I_D = 5$ mA. Then $r_d \approx 26$ mV / 5 mA = **5.2 Ω**. At 1 mA it would be 26 Ω, so the diode resists less when it conducts more.

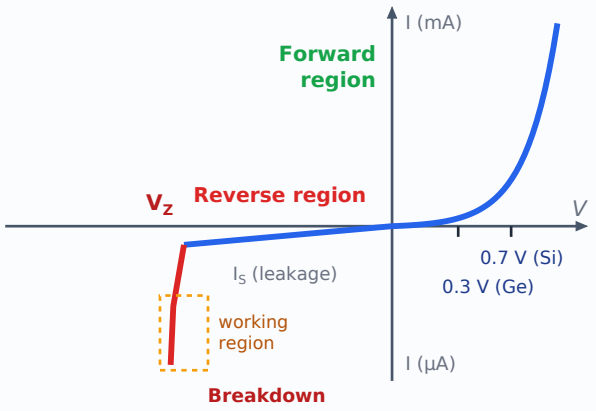
APPLICATIONS
Rectifiers (AC to DC), clippers and clampers, signal detectors, logic gates, reverse-polarity protection, voltage multipliers.

4 Zener Effect and Zener Diode

Zener diode
A specially **heavily doped** PN diode designed to work **safely in the reverse breakdown region**. In breakdown its voltage stays almost **constant (V_Z)** while the current changes widely, so it is a voltage reference/regulator.

Two breakdown mechanisms

	Zener effect	Avalanche effect
Occurs at	Low voltage (V_Z below ~5 V)	Higher voltage (above ~7 V)
Junction	Very heavy doping, very thin depletion layer	Lightly doped, wider depletion layer
Cause	Strong electric field (~10^7 V/m) pulls electrons straight out of covalent bonds (*field emission*)	Fast minority carriers knock out more electrons by *collision*, causing a chain (multiplication)
Temperature coefficient	Negative	Positive



Parameter	Meaning
V_Z	Zener (breakdown) voltage
$I_{Z(min)}$	Minimum current to stay in breakdown
$I_{Z(max)}$	Maximum safe current
$P_{Z(max)}$	$V_Z \times I_{Z(max)}$ (power rating)
Z_Z	Dynamic impedance $\Delta V_Z/\Delta I_Z$ (few Ω)

Fig 1.5 Zener diode characteristic and its main parameters

- **Forward bias:** behaves like an ordinary diode. **Reverse bias:** acts as an open circuit until V_Z, then conducts and holds V_Z.
- **Symbol:** diode symbol with the cathode bar bent like a "Z". Always use a **series resistor** to limit the current: $R_S = (V_{in} - V_Z) / (I_Z + I_L)$.

Example: series resistor for a Zener
Input = 12 V, V_Z = 5.1 V, load current I_L = 20 mA, minimum Zener current I_Z = 5 mA. R_S = (12 − 5.1) / (20 + 5) mA = 6.9 / 25 mA = **276 Ω** (use the nearest standard value, 270 Ω).

APPLICATIONS
Voltage regulators, reference voltage sources, over-voltage protection, waveform clipping, meter protection.

5 Bipolar Junction Transistor (BJT)

DEFINITION
A BJT is a **three-layer, two-junction, three-terminal** device (Emitter, Base, Collector) in which a **small base current controls a large collector current**. "Bipolar" because both electrons and holes take part in conduction.

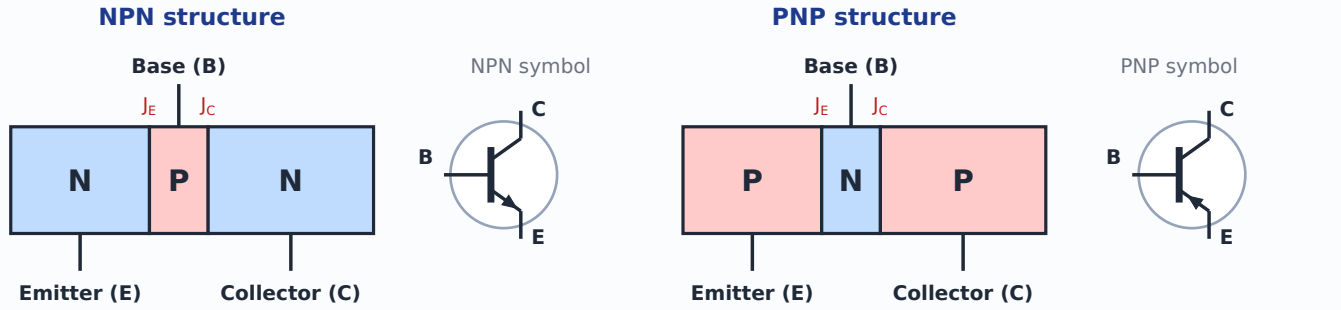


Fig 1.6 Structure and circuit symbols of NPN and PNP transistors (arrow shows direction of conventional emitter current)

Region	Doping	Size	Job
Emitter	Heavy	Medium	Emits (injects) majority carriers
Base	Light	**Very thin**	Controls the carrier flow
Collector	Moderate	**Largest**	Collects the carriers, dissipates heat

Working of an NPN transistor (active mode)

- Emitter-base junction is **forward biased**; collector-base junction is **reverse biased**.
- Electrons from the emitter enter the thin base; only ~1–5 % recombine (this is the small base current I_B).
- The rest are swept into the collector by the reverse field, giving a large $\mathbf{I_C}$.

Mode	EB junction	CB junction	Use
Cut-off	Reverse	Reverse	Switch OFF
Active	**Forward**	**Reverse**	**Amplifier**
Saturation	Forward	Forward	Switch ON

Current relations

$\mathbf{I_E = I_B + I_C}$

$\mathbf{\alpha = I_C / I_E}$ (0.95–0.99) | $\mathbf{\beta = I_C / I_B}$ (20–500)

$\mathbf{\beta = \alpha / (1 - \alpha)}$ | $\mathbf{\alpha = \beta / (1 + \beta)}$ | $\mathbf{I_E = (1 + \beta)\, I_B}$

Example

I_E = 10 mA and I_B = 0.2 mA. Then I_C = 10 − 0.2 = **9.8 mA**, α = 9.8/10 = **0.98**, β = 9.8/0.2 = **49** (check: 0.98/0.02 = 49 ✔).

6 Transistor Configurations: CB, CE and CC

A transistor has three terminals but a two-port network needs four, so **one terminal is made common** to input and output. The common terminal names the configuration.

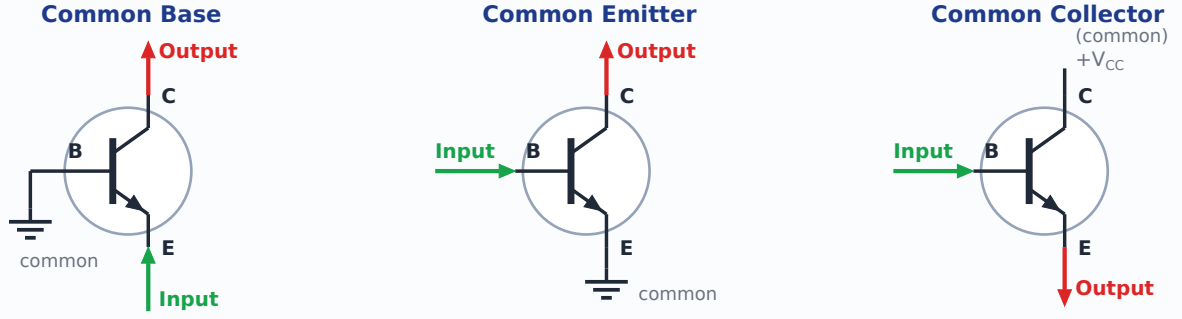


Fig 1.7 Three configurations (green = input, red = output, common terminal is grounded)

Property	CB	CE	CC (emitter follower)
Input terminal / Output	Emitter / Collector	Base / Collector	Base / Emitter
Input resistance	Very low (~50 Ω)	Medium (~1 kΩ)	Very high (~500 kΩ)
Output resistance	Very high (~1 MΩ)	High (~50 kΩ)	Very low (~50 Ω)
Current gain	α (just below 1)	β (high)	1 + β (high)
Voltage gain	High	High	Approx. 1 (slightly below)
Power gain	Medium	**Highest**	Medium
Phase (in to out)	0° (in phase)	**180° (inverted)**	0° (in phase)
Typical use	High-frequency / RF amplifiers	Audio amplifiers, switching	Buffer, impedance matching

EXAM TIP
CE is the most popular configuration because it gives high voltage, current *and* power gain. CC has gain ≈ 1 but is a perfect "buffer" between a high-impedance source and a low-impedance load.

7 Input and Output Characteristics

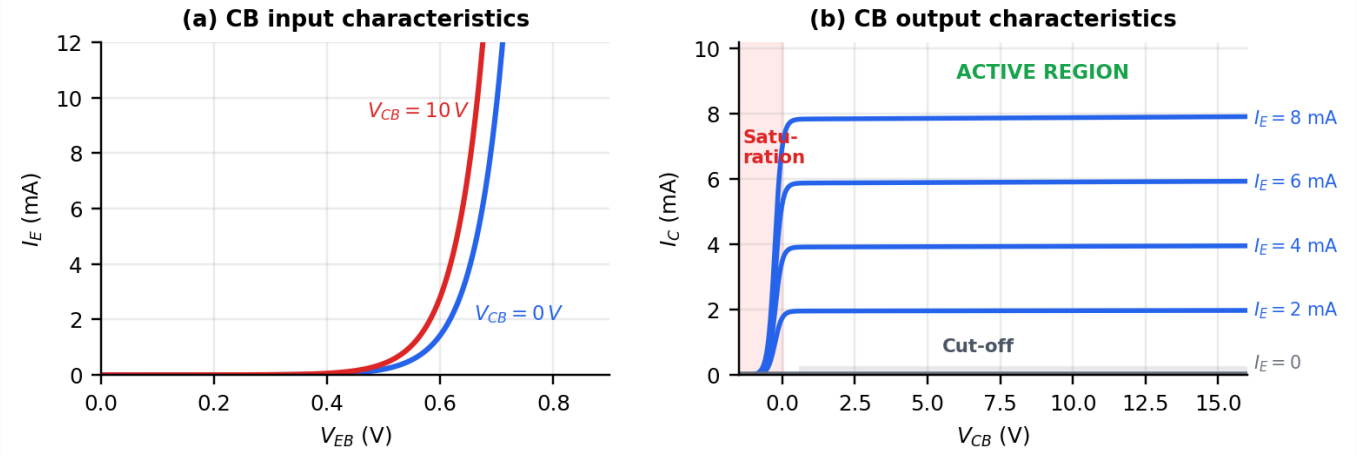


Fig 1.8(a) Common-base input and output characteristics

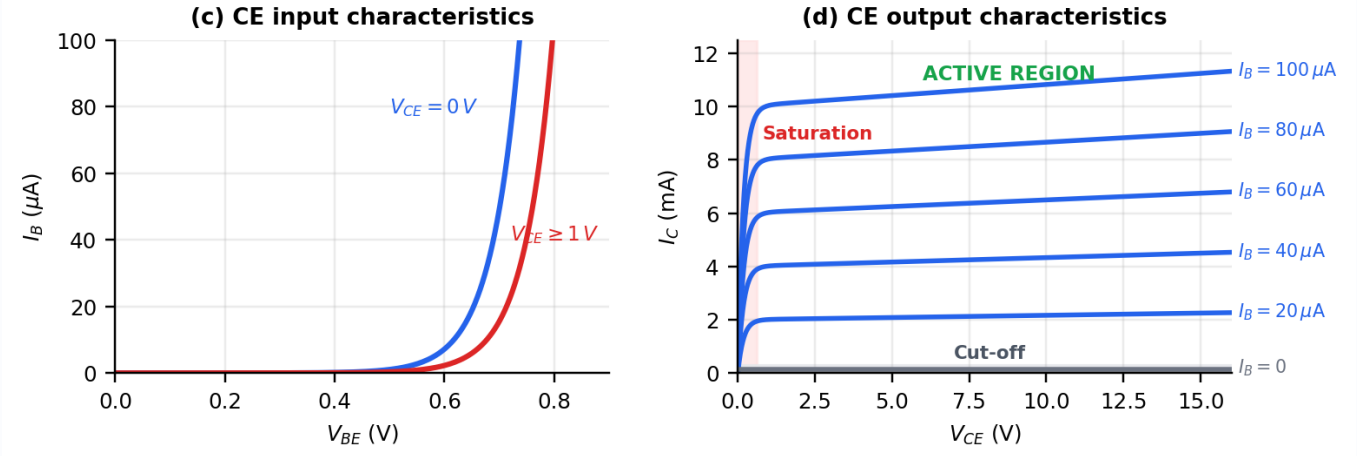


Fig 1.8(b) Common-emitter input and output characteristics

- **CB input (a):** I_E vs V_{EB} at constant V_{CB}. It looks like a forward diode curve; a larger V_{CB} shifts the curve slightly left.
- **CB output (b):** I_C vs V_{CB} at constant I_E. Almost flat lines, because $I_C \approx \alpha I_E$; the curve drops to zero only when V_{CB} is made forward (saturation).
- **CE input (c):** I_B vs V_{BE} at constant V_{CE}; a diode-like curve with a slightly higher cut-in at larger V_{CE}.

- **CE output (d):** I_C vs V_{CE} at constant I_B; $I_C \approx \beta I_B$ and the curves have a gentle upward slope.
- **CC characteristics:** input is I_B vs V_{BC}; output is I_E vs V_{EC}, which is nearly the same as the CE output (since $I_E \approx I_C$).

Region	CE condition	Meaning
Cut-off	$I_B = 0$, $I_C = I_{CEO}$ (tiny)	Transistor OFF; $V_{CE} \approx V_{CC}$
Active	EB forward, CB reverse	Linear amplification; $I_C = \beta I_B$
Saturation	$V_{CE(sat)} \approx 0.2$ V	Transistor fully ON; I_C is set by the external circuit

8 Small-Signal CE Amplifier (Elementary Treatment)

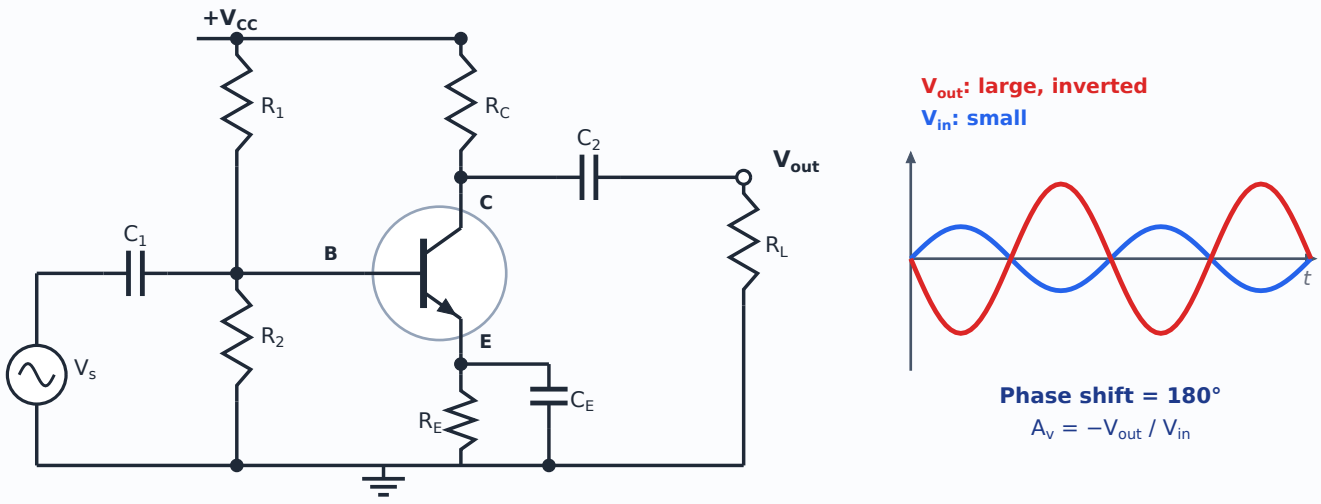


Fig 1.9 Voltage-divider-biased CE amplifier and its input/output waveforms

Role of each component

Component	Purpose
R_1, R_2	Voltage divider that sets a **stable DC bias** (Q-point) at the base
R_C	Converts the collector current change into an output voltage
R_E	Gives **stability** against temperature and β changes (negative feedback)
C_E (bypass)	Short-circuits R_E for AC so the gain is not reduced
C_1, C_2 (coupling)	Pass the AC signal and **block DC** between source, amplifier and load

Working

- **DC:** the biasing network fixes the **Q-point** in the middle of the active region.
- **AC:** in the +ve half-cycle of V_{in}, I_B and I_C increase, so the drop across R_C increases and V_{CE} **decreases**. In the −ve half-cycle the opposite happens.
- So the output is an **amplified and 180° phase-inverted** copy of the input.

Elementary small-signal analysis

DC: $V_B = V_{CC} \cdot R_2/(R_1+R_2)$; $V_E = V_B - 0.7$; $I_C \approx I_E = V_E/R_E$; $V_{CE} = V_{CC} - I_C(R_C+R_E)$

AC: $r_e = 26\text{ mV} / I_E$; $\mathbf{A_v = -(R_C \parallel R_L) / r_e}$; $Z_{in} \approx R_1 \| R_2 \| \beta r_e$; $Z_{out} \approx R_C$

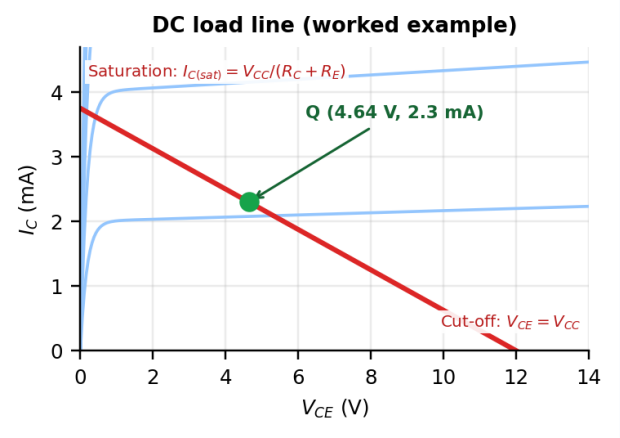
Worked example

$V_{CC} = 12$ V, $R_1 = 30$ kΩ, $R_2 = 10$ kΩ, $R_E = 1$ kΩ, $R_C = 2.2$ kΩ, $R_L = 10$ kΩ.

$V_B = 12 \times 10/40 = 3$ V → $V_E = 2.3$ V → $I_E \approx I_C = 2.3$ mA → $V_{CE} = 12 - 2.3 \times 3.2 =$ **4.64 V** (active ✔).

$r_e = 26/2.3 = 11.3$ Ω; $R_C \| R_L = 1.8$ kΩ; $\mathbf{A_v \approx -1800/11.3 \approx -160}$.

DC load line and Q-point



The **DC load line** joins the two extreme points of the circuit: **cut-off** ($I_C = 0$, $V_{CE} = V_{CC}$) and **saturation** ($V_{CE} \approx 0$, $I_C = V_{CC}/(R_C+R_E)$).

The **Q-point** (quiescent point) is the DC operating point on this line. It must sit near the **middle of the active region** so the signal swings equally up and down without clipping.

Q too high → *saturation clipping*; Q too low → *cut-off clipping*.

Fig 1.10 DC load line and Q-point for the worked example

Transistor as a switch

- **Cut-off** ($I_B = 0$): switch OPEN, $V_{CE} \approx V_{CC}$. • **Saturation** (large I_B): switch CLOSED, $V_{CE(sat)} \approx 0.2$ V.
- Used as the basic element of logic gates, relay/LED drivers and memory cells (links to Unit III).

9 Quick Summary and Exam Corner

Unit I in one glance

- Vacuum tube → transistor (1947) → IC (1958) → VLSI → nano-electronics: smaller, faster, cheaper, cooler.
- PN diode: forward = ON (knee 0.7 V Si, 0.3 V Ge); reverse = OFF with tiny leakage; breakdown at V_{BR}.
- Zener diode: operated in reverse breakdown, holds a constant V_Z; Zener effect (low V) and avalanche (high V).
- BJT: $I_E = I_B + I_C$; $\beta = \alpha/(1-\alpha)$. Modes: cut-off, active (amplifier), saturation.
- CE: highest power gain and 180° phase shift; CB: low R_{in}, good at high frequency; CC: gain ≈ 1, buffer.
- CE amplifier: R_1R_2 bias, R_E stability, C_E bypass, C_1C_2 coupling; $A_v = -R_C \| R_L / r_e$.

Short answer questions (2 marks)

1. What is the barrier potential of Si and Ge diodes?
2. Define dynamic resistance of a diode.
3. Distinguish Zener and avalanche breakdown (any two points).
4. Why is the base of a transistor made thin and lightly doped?
5. Define α and β and write the relation between them.
6. Why is CE the most widely used configuration?

Long answer questions (5 / 10 marks)

1. Explain the forward and reverse V-I characteristics of a PN junction diode with a neat diagram.
2. Explain the Zener diode characteristics and the Zener and avalanche breakdown mechanisms.
3. Draw the CE configuration and explain its input and output characteristics, marking the three regions.
4. Compare CB, CE and CC configurations in a table.
5. With a circuit diagram, explain the working of a small-signal CE amplifier. Derive A_v and state the role of each component.
6. Trace the evolution of electronics from vacuum tubes to nano-electronics.

Numerical practice (with answers)

1. A transistor has $I_B = 40$ μA and $\beta = 100$. Find I_C and I_E. *Ans: $I_C = 4$ mA, $I_E = 4.04$ mA*
2. Find β if $\alpha = 0.99$. *Ans: $\beta = 0.99/0.01 = 99$*
3. A Zener diode ($V_Z = 6.2$ V, $P_{Z(max)} = 0.5$ W). Find $I_{Z(max)}$. *Ans: 0.5/6.2 ≈ 80.6 mA*
4. For a CE amplifier, $I_E = 1.3$ mA, $R_C \| R_L = 2$ kΩ. Find A_v. *Ans: $r_e = 20\ \Omega$, $A_v = -100$*

Common mistakes to avoid

- Mixing up the biasing: **forward** = P to positive; **reverse** = P to negative.
- Saying “transistor amplifies current only”: CE amplifies **voltage, current and power**.
- Forgetting the **series resistor** with a Zener diode, or forgetting that CE **inverts** the signal.
- Drawing characteristic curves without labelling axes, constant parameter and the three regions.

Quick check: pick the right answer

1. The depletion region **widens** in: (a) forward bias (b) reverse bias (c) both (d) none. *Ans: (b)*
2. The phase shift of a CE amplifier is: (a) 0° (b) 90° (c) 180° (d) 360°. *Ans: (c)*
3. Emitter follower is another name for: (a) CB (b) CE (c) CC. *Ans: (c)*
4. A transistor used as an amplifier operates in the: (a) cut-off (b) active (c) saturation region. *Ans: (b)*
5. Zener breakdown below about 5 V is mainly due to: (a) collision (b) field emission (c) diffusion. *Ans: (b)*
6. Which configuration has the highest input resistance? (a) CB (b) CE (c) CC. *Ans: (c)*